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Online Book Market Analysis using Web Scraping and Exploratory Data Analysis

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Introduction

Problem Statement

The rapid growth of e-commerce has transformed the way people buy books. Online platforms provide a large variety of books with detailed information such as price, rating, availability, and category. Analyzing this data can help understand market trends, customer preferences, and pricing patterns.

Why this Project?

Thisproject was conducted to understand how real-world data can be collected and analyzed using webscraping and exploratory data analysis (EDA). Instead of using a pre-existing dataset, bookinformation such as price, category, and ratings was scraped directly from an online bookstore. This approach simulates real industry scenarios where analysts gather raw data from websites. The project helps identify patterns in book pricing, ratings, and category distribution.

Dataset Information

- Dataset : Books_Market_Analysis
- Source : books.toscrape.com
- Description : Contains scraped information about books including title, category, price, rating,

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DATA COLLECTION – API Based Web Scraping

API Scraping

- Collecting the data directly from the website provides a more realistic and practical dataset, similar to what data analysts and data engineers encounter in industry projects.
- Web scraping allows extracting publicly available information about books such as:

1. Book Title
2. Book Category
3. Price of the Book
4. Rating given by users
5. Availability (In stock / Out of stock
6. Product Description
7. Number of Reviews
8. Product Page URL

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Raw Dataset

1. The raw dataset was collected through web scraping from an online book store website.
2. It contains key information such as book title, price, rating, availability, and category.
3. The dataset required cleaning and preprocessing to handle missing values and convert data into proper formats for analysis.

Title	Category	Price	Availability	Rating	UPC	Price_excl_tax	Price_incl_tax	Number_of_reviews	URL
A Light in the Attic	Poetry	Â£51.77	In stock	Three	a897fe39b1053632	Â£51.77	Â£51.77	0	http://books.toscrape.com/catalogue/a-light-in...
Tipping the Velvet	Historical Fiction	Â£53.74	In stock	One	90fa61229261140a	Â£53.74	Â£53.74	0	http://books.toscrape.com/catalogue/tipping-th...
Soumission	Fiction	Â£50.10	In stock	One	6957f44c3847a760	Â£50.10	Â£50.10	0	http://books.toscrape.com/catalogue/soumission...
Sharp Objects	Mystery	Â£47.82	In stock	Four	e00eb4fd7b871a48	Â£47.82	Â£47.82	0	http://books.toscrape.com/catalogue/sharp-obje...
Sapiens: A Brief History of Humankind	History	Â£54.23	In stock	Five	4165285e1663650f	Â£54.23	Â£54.23	0	http://books.toscrape.com/catalogue/sapiens-a-...

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DataFrame Information

df.info() :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 10 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   title              500 non-null    object
 1   category           500 non-null    object
 2   price              500 non-null    object
 3   availability       500 non-null    object
 4   rating             500 non-null    object
 5   upc                500 non-null    object
 6   price_excl_tax     500 non-null    object
 7   price_incl_tax     500 non-null    object
 8   number_of_reviews  500 non-null    int64
 9   url                500 non-null    object
dtypes: int64(1), object(9)
```

DATA CLEANING

Data Quality Check

- The scraped datasetwasexamined for missing values, duplicate records, and inconsistent formats to ensure data accuracy and reliability. Data types and
- Column structures were verified to make the dataset suitable for exploratory data analysis and visualization.

Observation

- No missing values were detected across the dataset columns.
- No duplicate records were found, indicating the dataset is clean and reliable for analysis.

Insight

- Thescraped dataset is clean, consistent, and well-structured, allowing the analysis to proceed directly to feature engineering and exploratory data analysis (EDA) without significant preprocessing.

```
df.isnull().sum()
```

```
Title                 0
Category              0
Price                 0
Availability          0
Rating                0
UPC                   0
Price (excl. tax)     0
Price (incl. tax)     0
Description           0
URL                   0
dtype: int64
```

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FEATURE ENGINEERING

New features and transformations were applied to the scraped dataset to improve data quality and enable deeper analysis.These modifications help better understand book pricing patterns, rating distribution, and category trends within the online book market·

Category	Price	Availability	Rating	UPC	Price (excl. tax)	Price (incl. tax)	Description
Poetry	51.77	In stock	3	a897fe39b1053632	51.77	51.77	It's hard to imagine a world without A Light i...
Historical Fiction	53.74	In stock	1	90fa61229261140a	53.74	53.74	"Erotic and absorbing...Written with starling ...
Fiction	50.10	In stock	1	6957f44c3847a760	50.10	50.10	Dans une France assez proche de la nÃ´tre, un ...
Mystery	47.82	In stock	4	e00eb4fd7b871a48	47.82	47.82	WICKED above her hipbone, GIRL across her hear...
History	54.23	In stock	5	4165285e1663650f	54.23	54.23	From a renowned historian comes a groundbreaki...

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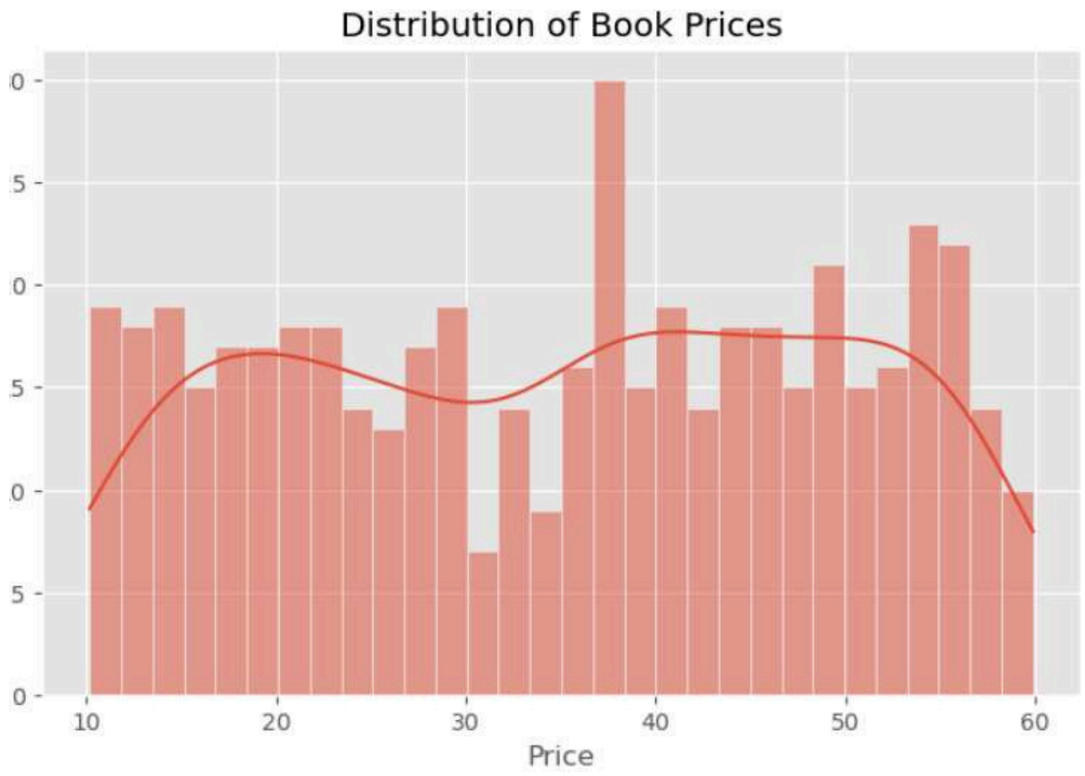
DATA VISUALIZATION

EDA – Univariate Analysis

What is the overal Idistribution of books prices in the dataset?

Observations:

1. It helps us understand how prices of books are spread across different ranges.
2. Most books fall into the medium price range rather than very high prices.
3. Only a few books have very high prices compared to the majority.
4. This graph helps identify the overall pricing pattern of books in the market.



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EDA – Univariate Analysis

How are book ratings distributed across different rating levels?

Observations:

- Mostbookshave ratings distributed across all categories from 1 to 5.
- Slightly higher counts are observed in lower rating categories, indicating mixed
- reader satisfaction.

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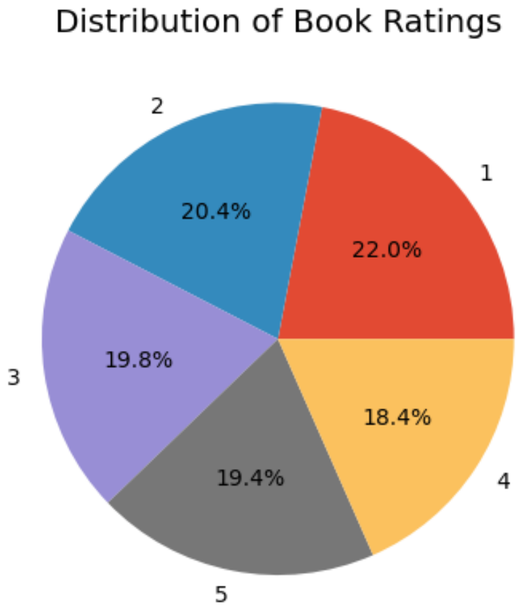
EDA – Univariate Analysis

What does the pie chart represent in the book dataset?



1. The pie chart shows the distribution of book ratings in the dataset.
2. Each slice represents the percentage of books for each rating category.
3. Larger slices indicate more books with that particular rating.
4. Most books fall in the higher rating categories like 3–5 stars.
5. This indicates that many books in the dataset received positive reader feedback.

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EDA – Bivariate Analysis

How does book rating vary with the price of books in the dataset?

Observations :

- Bookprices are spread across all rating levels from 1 to 5. Higher-rated books show a
- slightly higher median price compared to lower-rated books.

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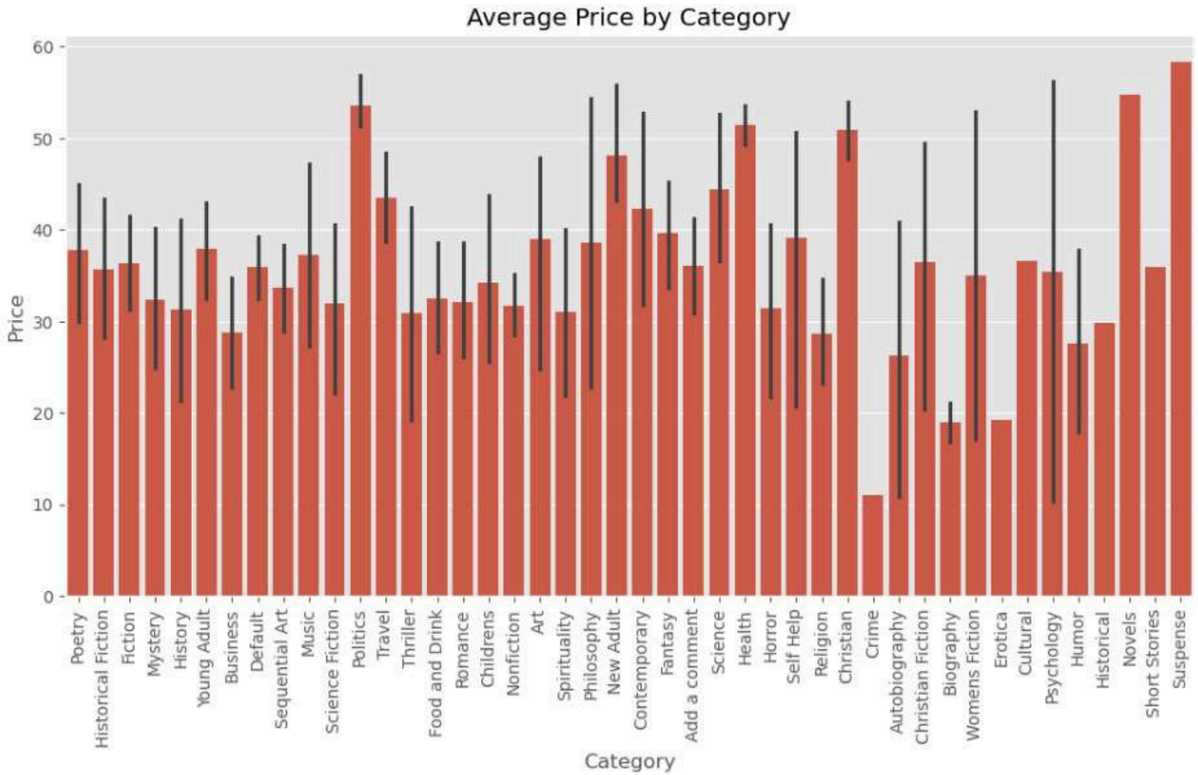
EDA – Bivariate Analysis

How does the average price of books vary across different categories?



Observations:

- The average price differs across book categories
- Some categories such as Suspense, Short Stories, and Health show relatively higher average prices compared to others.

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EDA – Bivariate Analysis

Is the any relationship between book availability and price?

Observations :

- Books that are in stock show a wide range of prices.
- Price values vary from low to high regardless of availability.

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EDA – Multivariate Analysis

How do price and rating vary based on the availability of books in the dataset?

Observations :

1. This analysis compares price, rating, and availability of books together.
2. Most available books have moderate prices and good ratings.
3. It shows that availability can influence both price and rating patterns.

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EDA – Multivariate Analysis

What relationship exists between the numerical variables in the dataset according to the correlation heatmap?

Observations:

1. The correlation heatmap shows the strength of relationship between numerical variables like price and rating.
2. Darker colors represent stronger correlations, while lighter colors indicate weaker relationships.
3. The heatmap helps quickly identify which variables are positively or negatively related in the dataset.

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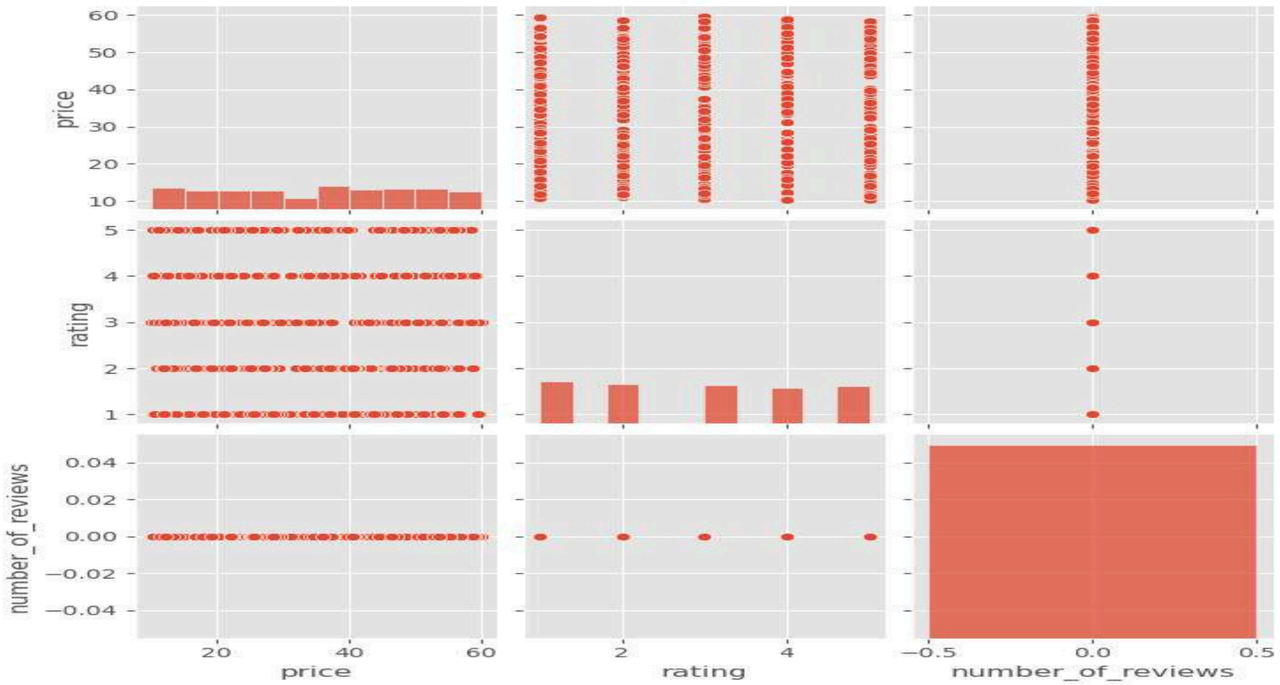
EDA – Multivariate Analysis

What relationships exist between price,rating,and number of reviews?



Observation:

- The pairplotvisualizes relationships between price, rating, and number of reviews simultaneously.
- The scatter plots show no strong visible linear relationship between these variables

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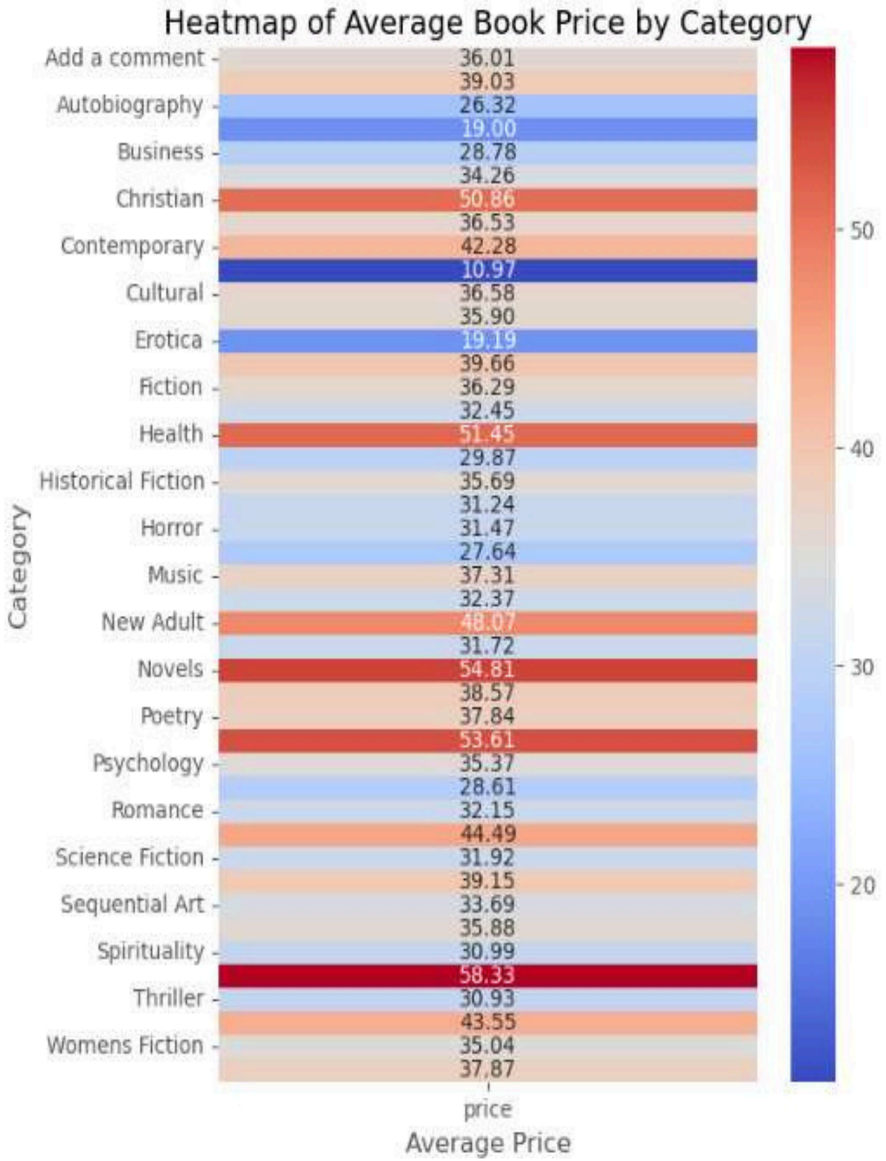
EDA –Heatmap Analysis

Which book categories have higher or lower average prices?

Observations:

- The shows variation iheatmapnaverage book prices across different categories.
- Categories such as Suspense, Novels, Politics, and Health have relatively higher average prices.



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Conclusion

1. The dataset shows that most books are priced in a moderate price range.
2. A majority of books have ratings between 3 and 5 stars, indicating good reader feedback.
3. Higher-rated books are more common in the dataset.
4. Price distribution analysis shows that very expensive books are fewer in number.
5. Bivariate analysis indicates that ratings and prices have a slight relationship.
6. Multivariate analysis shows that availability can influence price and rating patterns.
7. The correlation heatmap helps identify relationships between different variables in the dataset.
8. Overall, the analysis helps understand book pricing trends, ratings, and availability in the market.

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